

DataSet Introduction

This is a dataset: <https://www.kaggle.com/datasets/jacksondivakarr/car-crash-dataset?select=new+dataset.xlsx> I found on kaggle showing car crashes from the year 2003-2015 on car accidents across monroe county. It has datapoints on year, day, hour, collision type, injury type, primary reason for the accident, location of the accident, as well as the latitude and longitude of the accident.

Imported Tools

Tools that I imported were pandas, for data manipulation. numpy, for computations. matplotlib, for visualizations like bar charts and line charts. seaborn for visualizations like the correlation table that use heatmaps. and plotly express, for interactive graphs used in the map. as well as a multitude of scikit learns that I did not use however imported just incase.

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score
from sklearn.preprocessing import LabelEncoder

file_path = "C:/Users/19258/Downloads/new dataset.xlsx"
data = pd.read_excel(file_path)
display(data.head())
data.info()
```

	Year	Month	Day	Weekend?	Hour	Collision Type	Injury Type	Primary Factor	Reported_Location	Latitude	Longitude
0	2015	1	5	Weekday	0.0	2-Car	No injury/unknown	OTHER (DRIVER) - EXPLAIN IN NARRATIVE	1ST & FESS	39.159207	-86.525874
1	2015	1	6	Weekday	1500.0	2-Car	No injury/unknown	FOLLOWING TOO CLOSELY	2ND & COLLEGE	39.161440	-86.534848
2	2015	1	6	Weekend	2300.0	2-Car	Non-incapacitating	DISREGARD SIGNAL/REG SIGN	BASSWOOD & BLOOMFIELD	39.149780	-86.568890
3	2015	1	7	Weekend	900.0	2-Car	Non-incapacitating	FAILURE TO YIELD RIGHT OF WAY	GATES & JACOBS	39.165655	-86.575956
4	2015	1	7	Weekend	1100.0	2-Car	No injury/unknown	FAILURE TO YIELD RIGHT OF WAY	W 3RD	39.164848	-86.579625

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 53943 entries, 0 to 53942
Data columns (total 11 columns):
#   Column              Non-Null Count  Dtype
---  ---
0   Year                53943 non-null  int64
1   Month               53943 non-null  int64
2   Day                 53943 non-null  int64
3   Weekend?            53875 non-null  object
4   Hour                53718 non-null  float64
5   Collision Type       53937 non-null  object
6   Injury Type         53943 non-null  object
7   Primary Factor       52822 non-null  object
8   Reported_Location    53908 non-null  object
9   Latitude             53913 non-null  float64
10  Longitude            53913 non-null  float64
dtypes: float64(3), int64(3), object(5)
memory usage: 4.5+ MB
```

Question

The overarching question that I had was if there is a relation between any of these factors in causing car accidents, and which factors play the largest role. Furthermore how do these different aspects of car accidents relate to one another.

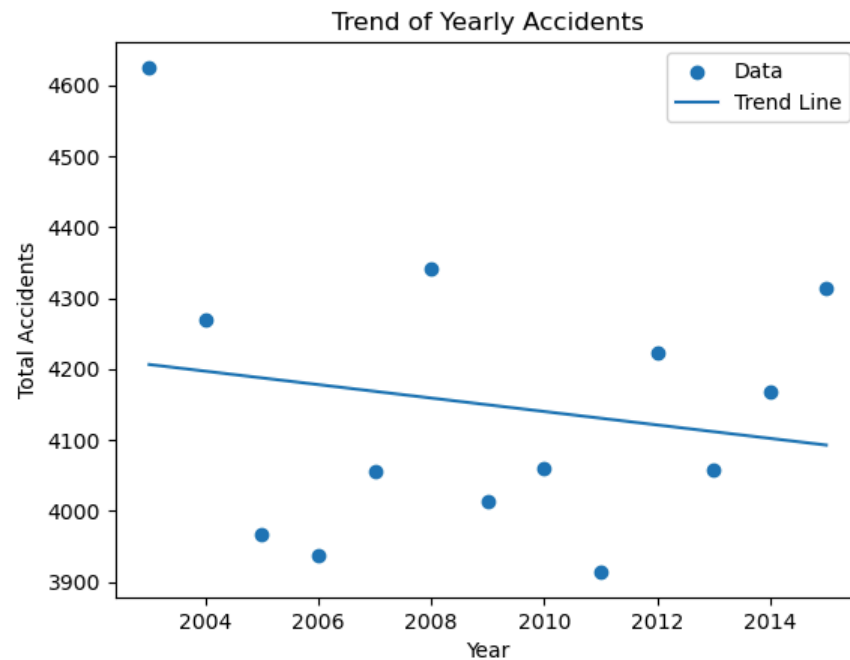
```
In [3]: accidents_per_year = data.groupby("Year").size().reset_index(name="Total Accidents")

X = accidents_per_year["Year"].values.reshape(-1, 1)
y = accidents_per_year["Total Accidents"].values

model = LinearRegression()
model.fit(X, y)

plt.scatter(X, y, label="Data")
plt.plot(X, model.predict(X), label="Trend Line")
plt.xlabel("Year")
plt.ylabel("Total Accidents")
plt.title("Trend of Yearly Accidents")
plt.legend()
plt.show()

print("Intercept:", model.intercept_)
print("Slope:", model.coef_[0])
```



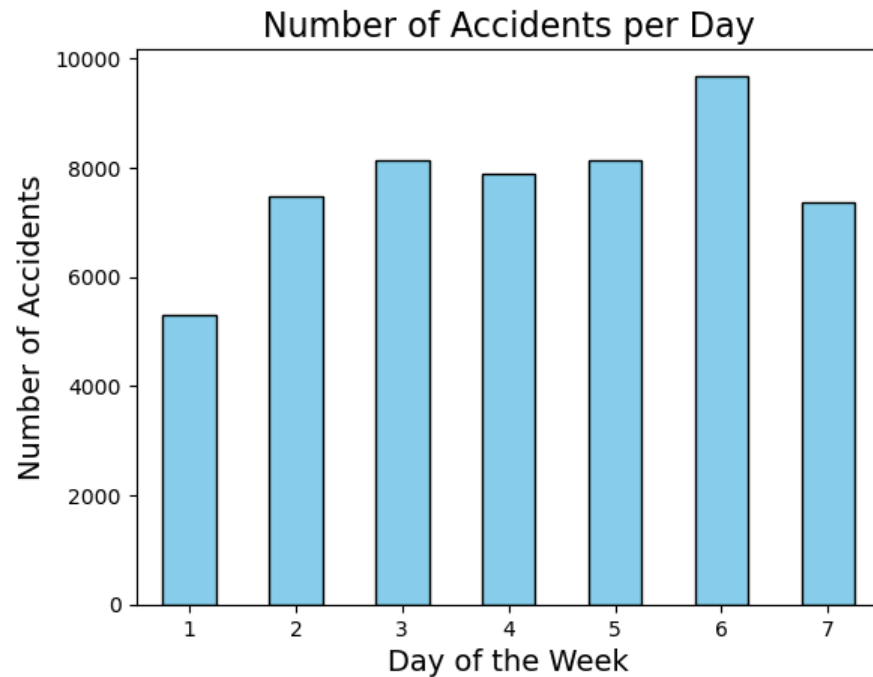
Intercept: 23168.73076923077
 Slope: -9.467032967032967

Trend of Yearly Accidents Analysis

This is a graph showing the trend of accidents across each year, using a linear regression model we see that it suggests a negative trend of around 9.47 however that's mainly due to the extreme outlier year in 2003, outside of that we can see that the accidents per year are generally within 500 accidents per year of each other with essentially random variation each year.

```
In [4]: accidents_per_day = data["Day"].value_counts().sort_index()

accidents_per_day.plot(kind="bar", color="skyblue", edgecolor="black")
plt.title("Number of Accidents per Day", fontsize=16)
plt.xlabel("Day of the Week", fontsize=14)
plt.ylabel("Number of Accidents", fontsize=14)
plt.xticks(rotation=0)
plt.show()
```



Number of Accidents per Day Analysis

This graph is showing the total number of accidents per day across 2003-2015, one thing about the dataset that is important to note is that day 1 is Monday. We can see that the day with the most accidents is Saturday and the day with the lowest is Monday. All the other days of the week are fairly similar in terms of number of accidents per day.

```
In [5]: total_accidents_per_day = data.groupby("Day").size()
average_accidents_per_day = total_accidents_per_day / data["Year"].nunique()

for day, avg_accidents in average_accidents_per_day.items():
    print(f"Day: {day}, Average Accidents: {avg_accidents:.2f}")
```

```
Day: 1, Average Accidents: 407.00  
Day: 2, Average Accidents: 574.85  
Day: 3, Average Accidents: 625.62  
Day: 4, Average Accidents: 606.15  
Day: 5, Average Accidents: 624.77  
Day: 6, Average Accidents: 744.62  
Day: 7, Average Accidents: 566.46
```

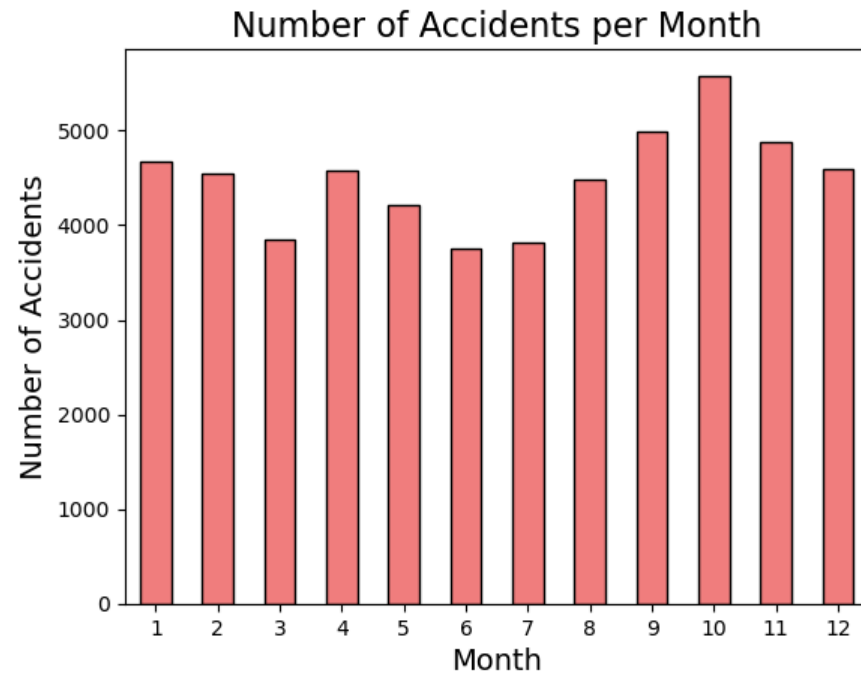
```
In [6]: average_accidents_1_to_5 = average_accidents_per_day.loc[1:5].mean()  
average_accidents_6_to_7 = average_accidents_per_day.loc[6:7].mean()  
  
print(f"Average Accidents (Weekdays): {average_accidents_1_to_5:.2f}")  
print(f"Average Accidents (Weekends): {average_accidents_6_to_7:.2f}")
```

```
Average Accidents (Weekdays): 567.68  
Average Accidents (Weekends): 655.54
```

Average Accidents Data

The above data is taking the data derived from the graph to get a better understanding of a per year basis. To help put an average on the above graph to get another way of understanding the graph. Furthermore I grouped the averages from above setting days 1-5 as the weekdays and days 6-7 as the weekends to see if there is any relation between the number of accidents on the weekday versus the weekend. It is clear that the weekend averages almost 90 more accidents than the weekdays, With the most accidents being on Saturday, and the least being on Monday.

```
In [7]: accidents_per_month = data["Month"].value_counts().sort_index()  
  
accidents_per_month.plot(kind="bar", color="lightcoral", edgecolor="black")  
plt.title("Number of Accidents per Month", fontsize=16)  
plt.xlabel("Month", fontsize=14)  
plt.ylabel("Number of Accidents", fontsize=14)  
plt.xticks(ticks=range(12), rotation=0)  
plt.show()
```

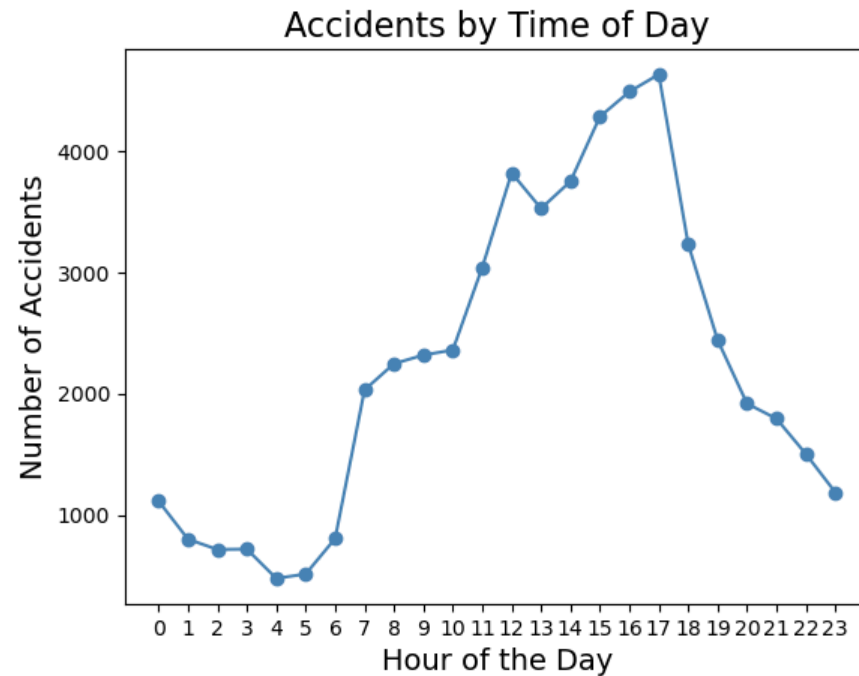


Month/Season Accident Analysis

This is a bar graph depicting the total number of accidents from 2003-2015 per month. We see that the months with the lowest accidents are March, June and July. We also see a rise start in August with a peak number of accidents in October. Generally we see that the fall seasons are responsible for a greater number of accidents in comparison to other seasons. This is most likely due to the poorer weather in the fall season.

```
In [8]: accidents_by_hour = data["Hour"].value_counts().sort_index()

plt.plot(accidents_by_hour.values, marker='o', color='steelblue')
plt.title("Accidents by Time of Day", fontsize=16)
plt.xlabel("Hour of the Day", fontsize=14)
plt.ylabel("Number of Accidents", fontsize=14)
plt.xticks(ticks=range(0, 24), labels=[str(i) for i in range(24)], rotation=0)
plt.show()
```

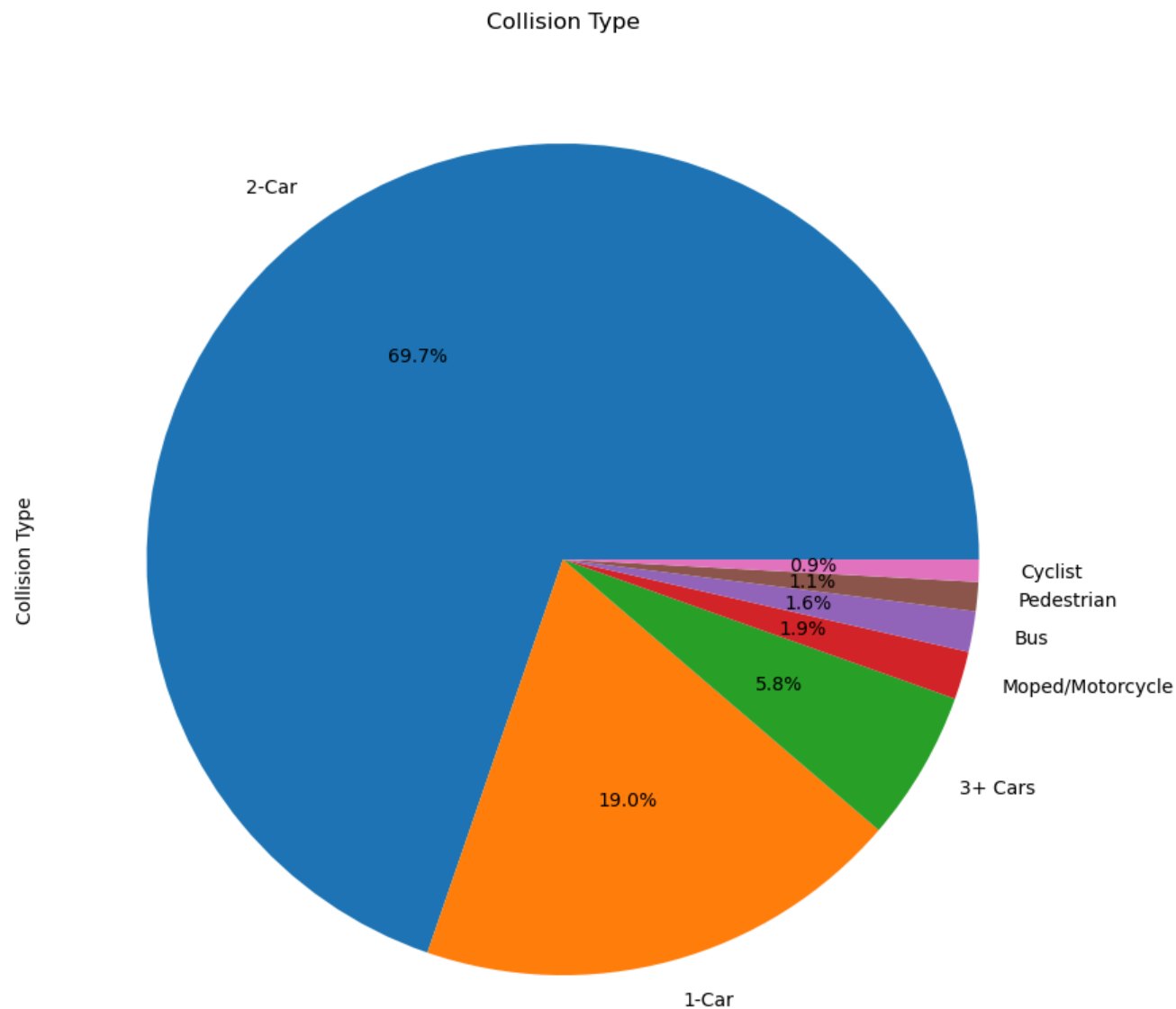


Analysis on Relation Between Time of Day and Accidents Frequency

This is a line graph showing the total number of accidents by the time of day from 2003-2015 in military time. We see that the lowest number of accidents happen at 4 in the morning however starting 6AM to 8AM there is a massive increase, as well as from 10AM to 12PM. after a slight decline to 13(1PM) from then to 17(5PM) there is a massive spike hitting its peak at 17(5). from 17(5) onwards there is a decline. This has a strong relation to human work times and rush hour as can see it is interesting to note that the evening traffic rush is associated with much more accidents than the morning traffic rush.

```
In [9]: collision_type_counts = data["Collision Type"].value_counts()

plt.figure(figsize=(10, 10))
collision_type_counts.plot.pie(autopct="%1.1f%%", startangle=0)
plt.title("Collision Type")
plt.show()
```



Most Common Accident Types

This is a pie chart showing the percentage of collision types from 2003-2015. We see that 2 car accidents are by far the most common accident type at 67.9 percent followed by 1 car at 19 percent and 3 car at 5.8 percent. Other accident types include, 3+ cars, moped, bus, pedestrian and motorcycle

```
In [4]: locations = data[['Latitude', 'Longitude', 'Reported_Location', "Hour", "Collision Type", "Injury Type", "Primary Factor"]].dropna()

fig = px.scatter_mapbox(
```

```

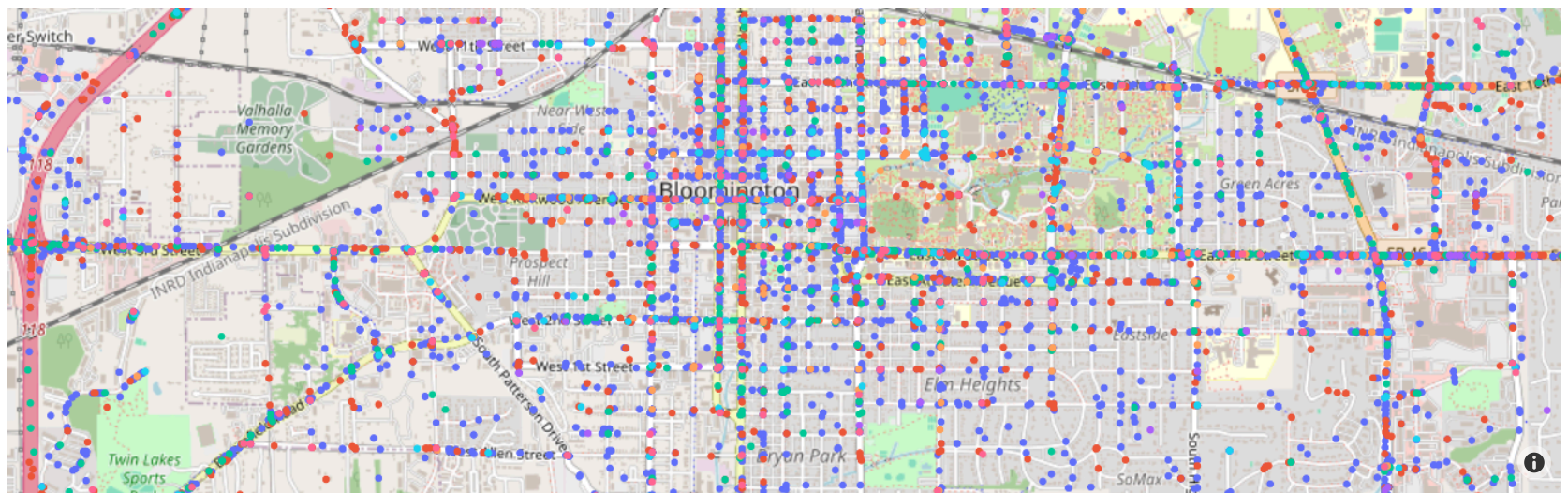
locations,
lat="Latitude",
lon="Longitude",
hover_name="Reported_Location",
hover_data={"Hour": True, "Injury Type": True, "Primary Factor": True, "Latitude": False, "Longitude": False},
color="Collision Type",
title="Interactive Map of Locations",
mapbox_style="open-street-map",
zoom=8,
)

fig.update_layout(
    mapbox=dict( center={"lat": 39.25, "lon": -86.45}),
    title="Accidents in Monroe County ",
    title_x=0.5
)

fig.show()

```

Accidents in Monroe County



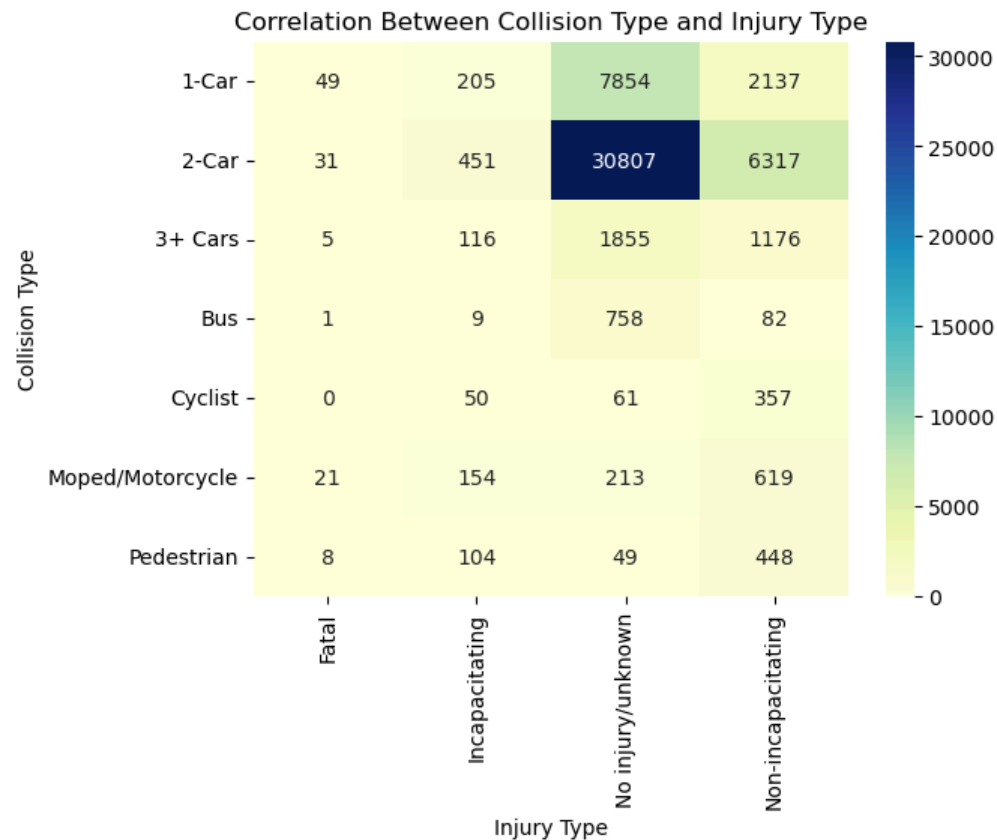
Accident Visualization

This is a map using longitude and latitude from every accident from 2003-2015 representing the location of every accident. Each accident is colorcoded so you know the collision type of the accident on the map. furthermore it also shows the injury type, primary factor as well as the hour in which the accident happened. This data is based around Bloomington Indiana and

we can see that most the accidents are happening along the major roads in the heart of the city rather than suburban areas.

```
In [15]: collision_injury_data = data[['Collision Type', 'Injury Type']]
correlation = pd.crosstab(collision_injury_data['Collision Type'], collision_injury_data['Injury Type'])

sns.heatmap(correlation, annot=True, fmt="d", cmap="YlGnBu")
plt.title('Correlation Between Collision Type and Injury Type')
plt.xlabel('Injury Type')
plt.ylabel('Collision Type')
plt.show()
```



Relation Between Collision Type and Injury Type

This is a heatmap of sorts that shows the relation between collision type and injury type mainly to get an understanding of what the relation between accident types and injury types are. we see the most common outcome is no injury across all accident types interesting to note is that motorcycles who make up 1.9 % of accidents have the highest fatality rate at 2.1% followed by pedestrians at 1.31%. Due to high frequency we see that 2 car collisions have the an extremley low ftality rate at 0.08%, the lowest is cyclist at 0%

```
In [19]: primary_factor_counts = data['Primary Factor'].value_counts()

primary_factor_table = pd.DataFrame({
    'Primary Factor': primary_factor_counts.index,
    'Number of Accidents': primary_factor_counts.values
})
```

```

})
print(primary_factor_table)

```

	Primary Factor	Number of Accidents
0	FAILURE TO YIELD RIGHT OF WAY	11193
1	FOLLOWING TOO CLOSELY	7359
2	OTHER (DRIVER) - EXPLAIN IN NARRATIVE	6158
3	UNSAFE BACKING	5188
4	RAN OFF ROAD RIGHT	2925
5	DISREGARD SIGNAL/REG SIGN	2206
6	SPEED TOO FAST FOR WEATHER CONDITIONS	1921
7	IMPROPER TURNING	1843
8	ANIMAL/OBJECT IN ROADWAY	1688
9	DRIVER DISTRACTED - EXPLAIN IN NARRATIVE	1656
10	UNSAFE SPEED	1499
11	ROADWAY SURFACE CONDITION	1270
12	LEFT OF CENTER	1078
13	IMPROPER LANE USAGE	985
14	ALCOHOLIC BEVERAGES	805
15	UNSAFE LANE MOVEMENT	756
16	OVERCORRECTING/OVERSTEERING	597
17	IMPROPER PASSING	496
18	OTHER (VEHICLE) - EXPLAIN IN NARRATIVE	472
19	OTHER (ENVIRONMENTAL) - EXPLAIN IN NARR	418
20	BRAKE FAILURE OR DEFECTIVE	361
21	PEDESTRIAN ACTION	292
22	DRIVER ASLEEP OR FATIGUED	267
23	DRIVER ILLNESS	182
24	VIEW OBSTRUCTED	175
25	CELL PHONE USAGE	141
26	NONE (DRIVER)	116
27	WRONG WAY ON ONE WAY	103
28	TIRE FAILURE OR DEFECTIVE	84
29	RAN OFF ROAD LEFT	60
30	PRESCRIPTION DRUGS	58
31	GLARE	53
32	ACCELERATOR FAILURE OR DEFECTIVE	50
33	INSECURE/LEAKY LOAD	46
34	OBSTRUCTION NOT MARKED	37
35	STEERING FAILURE	31
36	PASSENGER DISTRACTION	31
37	ILLEGAL DRUGS	29
38	OTHER TELEMATICS IN USE	28
39	OVERSIZE/OVERWEIGHT LOAD	26
40	ENGINE FAILURE OR DEFECTIVE	25
41	HEADLIGHT DEFECTIVE OR NOT ON	20
42	HOLES/RUTS IN SURFACE	15
43	TRAFFIC CONTROL INOPERATIVE/MISSING/OBSC	12
44	NONE (ENVIRONMENTAL)	12
45	NONE (VEHICLE)	11
46	OTHER LIGHTS DEFECTIVE	10
47	TOW HITCH FAILURE	8
48	ROAD UNDER CONSTRUCTION	7
49	JACKKNIFING	6
50	SEVERE CROSSWINDS	4
51	LANE MARKING OBSCURED	3
52	VIOLATION OF LICENSE RESTRICTION	3
53	SHOULDER DEFECTIVE	2
54	UTILITY WORK	1

Most Common Reason for Accidents

this shows the top reasons for accidents across 2003-2015 we see that failure to yield right of way is the most common by a large margin and utility work is the least common

```
In [38]: top_20_locations = data['Reported_Location'].value_counts().head(20)

top_20_locations_df = top_20_locations.reset_index()
top_20_locations_df.columns = ['Reported_Location', 'Count']

print(top_20_locations_df)
```

	Reported_Location	Count
0	E 3RD ST	375
1	W 3RD ST	222
2	SR37N & VERNAL	197
3	3RD ST	195
4	S WALNUT ST	172
5	E 10TH ST	153
6	N WALNUT ST	124
7	S COLLEGE MALL RD	123
8	SR37 & VERNAL	123
9	WALNUT ST	117
10	SR37S & VERNAL	113
11	SR37S & TAPP	112
12	3RD ST & JORDAN	108
13	10TH & COLLEGE AVE	105
14	EAST 3RD ST	104
15	E 17TH ST	102
16	SR37 & SR45	98
17	3RD ST & COLLEGE MALL	96
18	N FEE LN	94
19	13TH & INDIANA AVE	87

Most Common Accident Locations

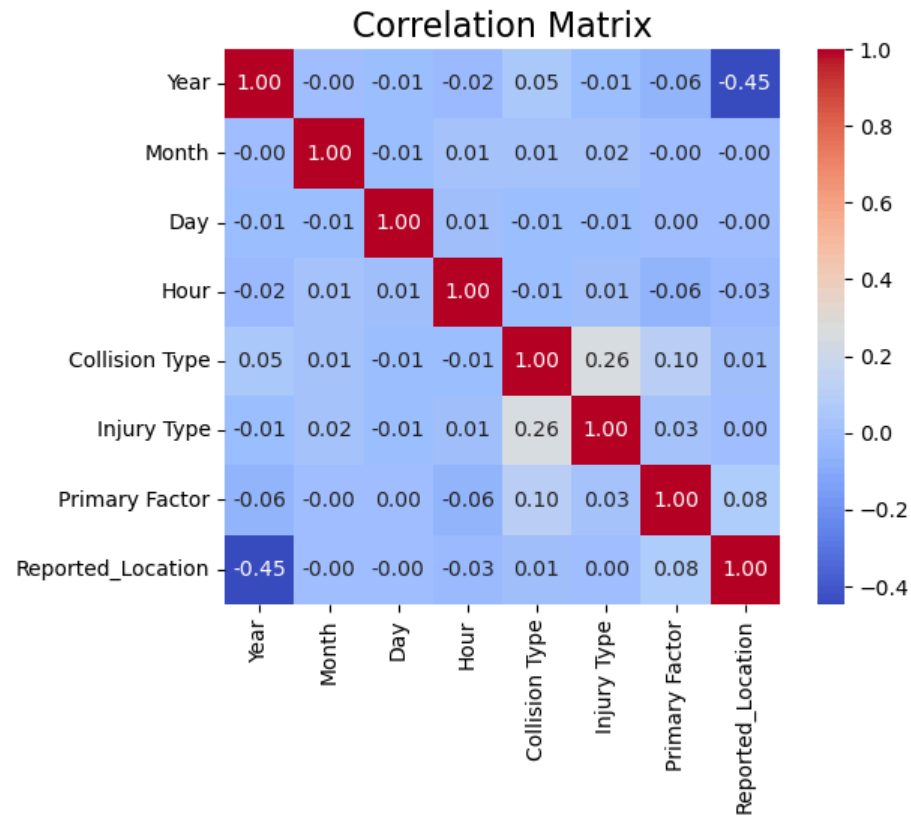
This shows the top 20 reported locations and the amount of accidents that have occurred, E 3rd street is the most followed by W 3rd street the least out of the top 20 is 13th and indiana avenue

```
In [20]: categorical_columns = ["Collision Type", "Injury Type", "Primary Factor", "Reported_Location"]
data[categorical_columns] = data[categorical_columns].apply(lambda col: pd.factorize(col)[0])

numeric_columns = ["Year", "Month", "Day", "Hour", "Collision Type", "Injury Type", "Primary Factor", "Reported_Location"]

updated_correlation_matrix = data[numeric_columns].corr()

sns.heatmap(updated_correlation_matrix, annot=True, fmt=".2f", cmap="coolwarm", cbar=True, square=True)
plt.title("Correlation Matrix", fontsize=16)
plt.show()
```



Final Takeaways

In summation we see there are factors that relate to increased accidents such as location, time of day, month, the day itself. However the relation between each factor is not as strong outside of collision type and injury type. With the possible exception of primary factor and collision type. My main takeaway from this is that the relation between car accidents is something that cannot be necessarily seen on a correlation, but rather something that has to be nuanced when observing data. Accidents are entirely on humans and each person is different. The Human aspect plays the single most important role as such as how drivers change the way the drive during certain seasons, rush hour. lack of understanding road signs, not paying attention to the road and so on. The only real thing that we can do to reduce accidents is be simply following the rules and by being a smart and defensive driver.

In []: